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B.tech CSE-A

Assignment - 2

- A-1
- 1) Entity Relationship Diagram
 - 2) Relational
 - 3) Object - Oriented
 - 4) Integrity

A-2 - Data Abstraction means hiding complex database details and showing only important information to users. In a healthcare system, it helps doctor, nurses & staff access patients records easily. There are 3 levels of data abstraction:

- 1) Physical Level: Describes how data is stored in the database. User don't interact with this level.
- 2) Logical level: Defines what data is stored, such as patient, doctor information and appointment and the relationship b/w them.
- 3) View Level: shows only required information to different users. for ex - doctors can see medical history, nurses can see patient vitals, and receptionist can see only appointment details.

Thus, data Abstraction makes the healthcare system Simple, secure and efficient.



A-3 Benefits:

- i) Improves data security and privacy
- ii) Makes the system easy to use for healthcare staff.
- iii) Provides data independence, so changes in database structures don't affect uses.
- iv) Helps in better data management.

Challenges:

- i) Designing abstraction layers can be complex.
- ii) It may cause slight performance overhead.
- iii) Implementation can be costly.
- iv) Staff may require training to use the system.

A-4 In a University, the main entities are student, course and Professor. These entities store important information about student, courses & teachers.

Entities & Attributes

i) Student

- Student - ID (Primary Key)

- Name

- Gender

- Age

- E-mail

ii) Course

- Course - ID (Primary Key)

- Course - Name

- Credits

- Departments



iii) Professor

- Professor-ID (Primary key)
- Professor-Name
- Department
- Specialization

Relationships

i) A student can enroll in many courses and a course can have ~~multiple~~ many students.

This is many-to-many relationship.

ii) A professor teaches courses and a course is taught by a professor.

This is a one-to-many relationship.

